

CSA-1610

Data Warehousing &
Data Mining

CO2 - AT2

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CSE (DS)

Q1)

a)

Data Mining: Data Mining is the process of extracting useful patterns, relationships and knowledge from large amounts of data using statistical, machine learning and database technique

Two Basic Data Mining tasks:

* Classification

* Clustering

KDD (Knowledge Discovery in Databases)

1. Data cleaning

* Removes missing, noisy and inconsistent data

2. Data Integration

* Combines data from multiple sources

3. Data Selection

* Selects relevant data for mining

4. Data Transformation

* Converts data into a suitable format

5. Data Mining

* Extracts useful patterns using algorithms

Pattern Evaluation

- * Identify Interesting and useful patterns
- 2. knowledge presentation
 - * presents the discovered knowledge using graphs, reports or visualization.

Roles of data preprocessing

- * Improve data quality
- * Remove duplicate and noisy data
- * Handle missing values
- * Reduce data size.
- * Increases mining accuracy.

Q1 b) Data Cleaning is the process of detecting and correcting incomplete, inaccurate, duplicate or inconsistent data before data mining.

Two common Data Quality issues : 1. Missing Values 2. Noisy data

Handling Missing values

- Ignore the record
- 1. Fill manually
- 2. Replace with mean
- 3. Replace with median
- 4. Replace with most freq value
- 5. Predict using ML

Ex : Age : 20, 22, ..., 25

$$\text{Mean } \frac{(20+22+25)}{3} = 22.33$$

∴ 20, 22, 22.33, 25

Handling noisy Data

1. Binning
2. Regression
3. clustering

Ex : Marks : 55, 56, 57, 120, 58.

120 is an outlier

- * Replace it using smoothing techniques or remove it.

Q2 a)

Data Integration: Data integration is the process of combining data from multiple source into single, consistent dataset.

Two challenges.

1. Data redundancy
2. Data inconsistency

Handling redundancy.

- * Remove duplicate records
- * Use primary keys and Correlation analysis.

Handling inconsistency

- * Standardize data formats.
- * Resolve conflicting values using validation rules.

Q2 b)

* Covariance analysis measure how two variable change together.

- positive Covariance: variable increase or decrease together
- Negative Covariance: One variable increase while other decrease.
- Zero Covariance: variables are independent.

Two covariance formulae

1. population Covariance:
$$\text{Cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

2. sample Covariance:
$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Q3 a)

Stock A (x)	Stock B (y)
2	5
3	8
5	10
4	11
6	14

Mean

$$\bar{x} = \frac{2+3+5+4+6}{5}$$

$$\bar{x} = 4$$

$$\bar{y} = \frac{5+8+10+11+14}{5}$$

$$\bar{y} = 9.6$$

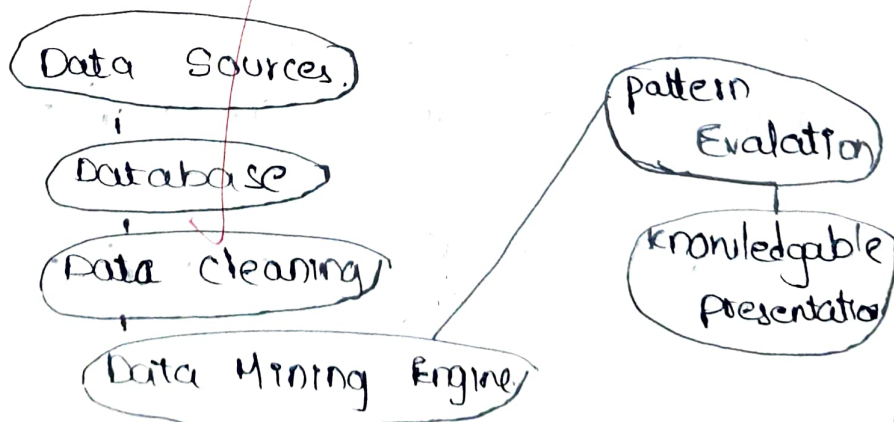
Calculate Covariance

X	Y	$x - \bar{x}$	$y - \bar{y}$	product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

$$\sum (x - \bar{x})(y - \bar{y}) = 20$$

$$\therefore \text{cov}(x, y) = \frac{20}{5-1} = 5$$

b) Architecture of Typical Data Mining System



Components

- * Data Source
- * Data warehouse
- * Data Mining Engine
- * Knowledge presentation

Q4) Given,

a) $X = \{4, 5, 6, 7, 9\}$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean } \bar{X} = \frac{4+5+6+7+9}{5} = 6.2$$

$$\bar{Y} = \frac{8+3+7+5+6}{5} = 5.8$$

Calculating Table.

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	Product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.5	0.2	0.56

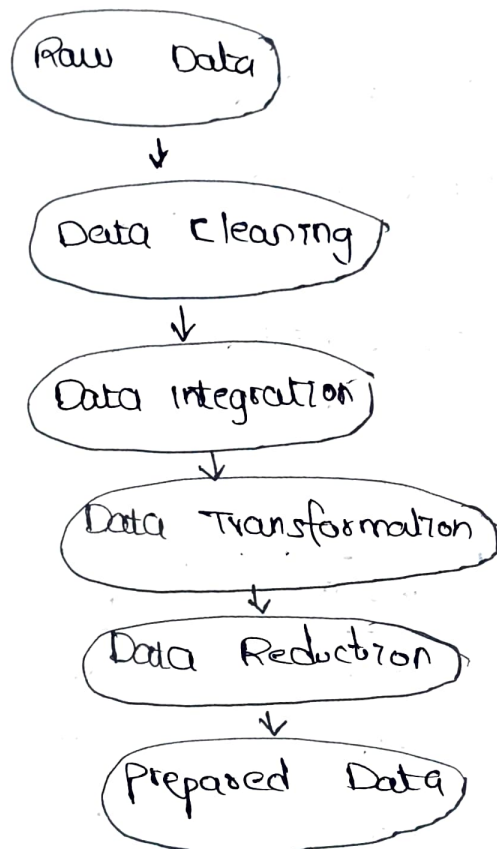
$$\sum (X - \bar{X})(Y - \bar{Y}) = -1.80$$

$$\text{COV}(X, Y) = \frac{-1.80}{5-1} = \frac{-1.80}{4} = -0.45$$

b) Major steps in data preprocessing

1. Data cleaning - Remove missing and noisy data
2. Data integration - combines data from multiple sources
3. Data Transformation - Converts data into suitable format
4. Data Reduction - Reduces data size while preserving

preprocessing Pipeline:



- cleaning removes missing values and outliers
- integration combines different data sources
- Transformation Standardize data
- Reduction decreases storage & processing time

Q5) a)

Equal Frequency Binning

Given data :

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30,
33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. $\Rightarrow 27$ values

Bin 1

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2

22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3

35, 35, 35, 36, 40, 45, 46, 52, 70.

$\Rightarrow$ Equal frequency binning groups the data into bins containing same number of value which helps smooth noisy data while preserving the distributions.

b) Data Discretization : Data discretization is the process of converting continuous data into discrete interval

Importance :

- * Reduce data size
- * Remove noise
- * Improving mining efficiency
- * Makes data easier to analyse.

Equal width Binning

* Divides data into interval of equal width.

$$\text{width} = (\text{Max} - \text{Min}) / \text{no. of bins}$$

Example :

Data : 10, 20, 30, 40, 50

Bins :

* 10 - 23

* 24 - 27

* 38 - 50

Equal frequency Binning :

* Divides data so that each bin has an equal no. of values

Example :

Data : 10, 20, 30, 40, 50, 60

Bin :

Bin 1 : 10, 20, 30

Bin 2 : 40, 50, 60

Impact On Data Analysis :

* Equal - width : Suitable for uniformly distributed data

* Equal - Frequency : Better for skewed data and reduces the effect of outliers.